

LAPORAN PRAKTIKUM WORKSHOP WEB DASAR

Javascript DOM



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BAB I

PENDAHULUAN

1. Latar Belakang

Lahirnya DOM berakar dari "Perang Browser" pada akhir 1990-an antara Netscape Navigator dan Microsoft Internet Explorer. Setiap browser menggunakan metode internalnya sendiri untuk berinteraksi dengan elemen web, sehingga pengembang kesulitan membuat situs web yang berfungsi baik di berbagai platform

2. Identifikasi Masalah

- Performa Lambat (Render-Blocking & Reflow)
- Celah Keamanan XSS (Cross-Site Scripting) berbasis DOM
- Masalah Skrip dan SEO (Crawl & Index)
- Kesalahan Navigasi dan Null Reference

3. Rumusan masalah

- Fokus pada Implementasi & Dasar
- Fokus pada Interaksi & Peristiwa (Events)
- Fokus pada Performa & Penerapan (Tingkat Lanjut)

BAB II

LANDASAN TEORI

1. Pengertian Javascript DOM

Javascript DOM (Document Object Model) adalah antarmuka pemrograman (API) yang merepresentasikan halaman web dalam bentuk struktur pohon hierarkis. DOM memungkinkan bahasa pemrograman seperti JavaScript untuk mengakses, mengubah, menambah, atau menghapus elemen, konten, struktur, dan gaya dokumen HTML atau XML secara dinamis.

2. Struktur Javascript DOM

- Document Node: Titik akar (root) dari seluruh pohon DOM.
- Element Node: Mewakili tag HTML seperti `<body>`, `<h1>`, `<p>`, dan `<div>`.
- Text Node: Mewakili isi text di dalam element HTML.
- Attribute Node: Mewakili atribut dari sebuah elemen (seperti `class`, `id`, `href`).

3. Fungsi Javascript DOM

Sebagai antarmuka pemrograman yang menjembatani bahasa javascript dengan dokumen HTML atau XML. Hal ini memungkinkan Javascript untuk mengakses, mengubah, menambahkan, atau menghapus element HTML, teks, maupun atribut untuk membangun halaman web yang interaktif.

BAB III

Paktikkum Dasar Tailwind.Css Dasar

1. Alat dan Bahan:

- Laptop
- Visual Studio Code
- Git
- GitHUb

2. Proses Pembuatan Laporan : Aplikasi Todo List Lengkap

Langkah 1 Buat folder Tugas pada modul -6

```
#Pastikan berada pada repository kamu
Mkdir modul-6
Cd modul-6
Mkdir tugas
```

Langkah 2 Buat file tugas index.html, tugas/style.css,dan tugas/script.js

```
Cd tugas
Touch index.html style.css script.js
```

Langkah 3 Isi code ini pada file index.html

```
<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-
scale=1.0" />
```

```

<title>Toko Komputer Online - Keranjang Belanja</title>
<link rel="stylesheet" href="style.css" />
</head>
<body>
  <main class="page">
    <section class="heading">
      <div>
        <h1>🖥️ Toko Komputer Online</h1>
        <p>Belanja perangkat komputer berkualitas tinggi dengan
harga terjangkau. Tambahkan produk ke keranjang, atur jumlah
pembelian, dan checkout dengan mudah.</p>
      </div>
      <div class="badge" id="iconBadge">0 item</div>
    </section>

    <div class="layout">
      <section>
        <div class="product-grid" id="productGrid" aria-
live="polite"></div>
      </section>

      <aside class="cart-panel" aria-label="Ringkasan keranjang
belanja">
        <div class="cart-header">
          <div>
            <h2>Keranjang Anda</h2>
            <p id="cartStatus">Belum ada produk dalam keranjang.</p>
          </div>
          <div class="badge" id="cartCount">0</div>
        </div>

        <ul class="cart-list" id="cartList"></ul>

        <div class="summary">
          <div class="summary-row">
            <span>Total item</span>
            <span id="summaryItems">0</span>
          </div>
          <div class="summary-row total">
            <span>Harga total</span>
            <span id="summaryTotal">Rp 0</span>
          </div>
        </div>

        <button class="checkout" id="checkoutButton"
type="button">Checkout</button>
      </aside>
    </div>
  </main>

```

```
<script src="script.js"></script>
</body>
</html>
```

Langkah 4 Isi code ini pada file style.css

```
:root {
  color-scheme: light;
  font-family: Inter, system-ui, sans-serif;
  font-size: 16px;
  color: #1f2937;
  background: #f0f4f8;
}

* {
  box-sizing: border-box;
}

body {
  margin: 0;
  min-height: 100vh;
  background: linear-gradient(180deg, #f0f4f8 0%, #e8ecf1 100%);
}

.page {
  max-width: 1200px;
  margin: 0 auto;
  padding: 24px;
  display: grid;
  gap: 24px;
}

.heading {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 16px;
  flex-wrap: wrap;
}

.heading h1 {
  margin: 0;
  font-size: clamp(1.8rem, 2.4vw, 2.4rem);
  letter-spacing: -0.03em;
}
```

```
.heading p {
  margin: 0;
  color: #475569;
  max-width: 700px;
  line-height: 1.5;
}

.layout {
  display: grid;
  grid-template-columns: 1.6fr 1fr;
  gap: 24px;
}

.product-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
  gap: 20px;
}

.card {
  border-radius: 20px;
  overflow: hidden;
  background: #ffffff;
  box-shadow: 0 16px 45px rgba(15, 23, 42, 0.08);
  display: flex;
  flex-direction: column;
}

.card img {
  width: 100%;
  aspect-ratio: 4 / 3;
  object-fit: cover;
  display: block;
}

.card-body {
  padding: 18px;
  display: grid;
  gap: 10px;
  flex: 1;
}

.card-title {
  font-size: 1.1rem;
  margin: 0;
  font-weight: 700;
  color: #111827;
}
```



```
.card-price {
  margin: 0;
  color: #1e40af;
  font-weight: 700;
}

.card-actions {
  margin-top: auto;
  display: flex;
  justify-content: space-between;
  align-items: center;
  gap: 12px;
}

.button,
.icon-button {
  border: none;
  border-radius: 12px;
  cursor: pointer;
  font-weight: 700;
  transition: transform 0.15s ease, background 0.15s ease, box-
shadow 0.15s ease;
}

.button {
  background: #1e40af;
  color: white;
  padding: 0.8rem 1rem;
  flex: 1;
}

.button:hover {
  transform: translateY(-1px);
  box-shadow: 0 12px 28px rgba(30, 64, 175, 0.18);
}

.cart-panel {
  position: sticky;
  top: 24px;
  align-self: start;
  background: #ffffff;
  border-radius: 24px;
  padding: 24px;
  box-shadow: 0 16px 45px rgba(15, 23, 42, 0.08);
  display: grid;
  gap: 18px;
}

.cart-header {
```

```
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 12px;
}

.cart-header h2 {
    margin: 0;
    font-size: 1.2rem;
}

.badge {
    display: inline-flex;
    min-width: 2.2rem;
    padding: 0.5rem 0.7rem;
    border-radius: 999px;
    background: #dbeafe;
    color: #1e40af;
    font-weight: 700;
    justify-content: center;
}

.cart-list {
    display: grid;
    gap: 16px;
    margin: 0;
    padding: 0;
    list-style: none;
}

.cart-item {
    display: grid;
    grid-template-columns: 1fr auto;
    gap: 12px;
    padding: 14px;
    border-radius: 18px;
    background: #f8fafc;
    border: 1px solid #e2e8f0;
}

.item-info {
    display: grid;
    gap: 4px;
}

.item-info strong {
    font-size: 0.98rem;
}
```

```
.item-subtitle {
  color: #475569;
  font-size: 0.95rem;
}

.item-actions {
  display: grid;
  gap: 10px;
  justify-items: end;
  min-width: 108px;
}

.qty-controls {
  display: inline-flex;
  align-items: center;
  gap: 6px;
  border-radius: 999px;
  background: white;
  border: 1px solid #cbd5e1;
  padding: 4px;
}

.icon-button {
  width: 34px;
  height: 34px;
  background: #e2e8f0;
  color: #0f172a;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  font-size: 1rem;
}

.icon-button:hover {
  background: #cbd5e1;
}

.remove-button {
  background: transparent;
  color: #ef4444;
  padding: 0;
  min-height: 0;
  font-size: 0.95rem;
}

.summary {
  display: grid;
  gap: 12px;
  padding-top: 12px;
}
```

```
border-top: 1px solid #e2e8f0;
}

.summary-row {
  display: flex;
  justify-content: space-between;
  gap: 12px;
  color: #475569;
}

.summary-row.total {
  font-weight: 700;
  color: #111827;
}

.checkout {
  width: 100%;
  padding: 0.95rem 1rem;
  border-radius: 14px;
  border: none;
  background: #1e40af;
  color: white;
  font-size: 1rem;
  font-weight: 700;
  cursor: pointer;
}

.checkout:disabled {
  opacity: 0.55;
  cursor: not-allowed;
}

.empty-state {
  color: #64748b;
  padding: 18px 16px;
  border-radius: 18px;
  background: #f8fafc;
  border: 1px dashed #cbd5e1;
  text-align: center;
}

@media (max-width: 900px) {
  .layout {
    grid-template-columns: 1fr;
  }

  .cart-panel {
    position: relative;
    top: 0;
  }
}
```

```
}  
}
```

Langkah 5 Isi code ini pada file Script.js

```
const products = [  
  {  
    id: 1,  
    name: 'RAM DDR4 16GB 3200MHz',  
    price: 450000,  
    image: 'https://images.unsplash.com/photo-1611855719306-855681466066?q=80&w=1119&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D'  
  },  
  {  
    id: 2,  
    name: 'SSD 512GB NVMe PCIe 4.0',  
    price: 550000,  
    image: 'https://images.unsplash.com/photo-1588259341633-32e9693ccb59?q=80&w=1073&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D'  
  },  
  {  
    id: 3,  
    name: 'Monitor LED 24 Inch Full HD',  
    price: 1200000,  
    image: 'https://plus.unsplash.com/premium_photo-1669380425564-6e1a281a4d30?q=80&w=1170&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D'  
  },  
  {  
    id: 4,  
    name: 'Mechanical Keyboard RGB',  
    price: 750000,  
    image: 'https://images.unsplash.com/photo-1626958390943-a70309376444?q=80&w=1170&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D'  
  },  
  {  
    id: 5,  
    name: 'Wireless Mouse Gaming',  
    price: 350000,  
    image: 'https://images.unsplash.com/photo-1629121291243-7b5e885cce9b?q=80&w=764&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D'  
  },  
]
```

```
    id: 6,
    name: 'Headphone Gaming Stereo',
    price: 650000,
    image: 'https://images.unsplash.com/photo-1677086813101-496781a0f327?q=80&w=1074&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D'
  },
  {
    id: 7,
    name: 'USB Hub 7-Port 3.0',
    price: 280000,
    image: 'https://images.unsplash.com/photo-1760376789478-c1023d2dc007?q=80&w=1170&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D'
  },
  {
    id: 8,
    name: 'Power Supply 650W Modular',
    price: 1100000,
    image: 'https://m.media-amazon.com/images/I/71HEQ9FBF9L._AC_SL1500_.jpg'
  }
];

let cart = [];

const productGrid = document.getElementById('productGrid');
const cartList = document.getElementById('cartList');
const cartCount = document.getElementById('cartCount');
const iconBadge = document.getElementById('iconBadge');
const cartStatus = document.getElementById('cartStatus');
const summaryItems = document.getElementById('summaryItems');
const summaryTotal = document.getElementById('summaryTotal');
const checkoutButton = document.getElementById('checkoutButton');

function formatCurrency(value) {
  return new Intl.NumberFormat('id-ID', {
    style: 'currency',
    currency: 'IDR',
    maximumFractionDigits: 0
  }).format(value);
}

function renderProducts() {
  productGrid.innerHTML = products.map(product => {
    return `
      <article class="card">
        
        <div class="card-body">
```

```
        <h3 class="card-title">${product.name}</h3>
        <p class="card-price">${formatCurrency(product.price)}</p>
        <div class="card-actions">
            <span>${formatCurrency(product.price)}</span>
            <button class="button" type="button" data-action="add"
data-id="${product.id}">Tambah ke Keranjang</button>
        </div>
    </div>
</article>
`
    }).join('');
}

function getCartItemCount() {
    return cart.reduce((sum, item) => sum + item.quantity, 0);
}

function getCartTotal() {
    return cart.reduce((sum, item) => {
        const product = products.find(product => product.id ===
item.productId);
        return sum + (product.price * item.quantity);
    }, 0);
}

function renderCart() {
    const totalItems = getCartItemCount();
    const totalPrice = getCartTotal();

    cartCount.textContent = totalItems;
    iconBadge.textContent = `${totalItems} item`;
    summaryItems.textContent = totalItems;
    summaryTotal.textContent = formatCurrency(totalPrice);
    cartStatus.textContent = totalItems === 0 ? 'Belum ada produk
dalam keranjang.' : 'Item sudah ditambahkan ke keranjang.';
    checkoutButton.disabled = totalItems === 0;

    if (cart.length === 0) {
        cartList.innerHTML = `
            <li class="empty-state">Keranjang kosong. Tambahkan produk
terlebih dahulu.</li>
        `;
        return;
    }

    cartList.innerHTML = cart.map(item => {
        const product = products.find(product => product.id ===
item.productId);
        return `
```

```

        <li class="cart-item">
          <div class="item-info">
            <strong>${product.name}</strong>
            <span class="item-subtitle">${formatCurrency(product.price)}
            x
            ${item.quantity}</span>
            <span class="item-subtitle">Subtotal:
            ${formatCurrency(product.price * item.quantity)}</span>
          </div>
          <div class="item-actions">
            <div class="qty-controls">
              <button class="icon-button" type="button" data-
              action="decrease" data-id="${item.productId}">-</button>
              <span>${item.quantity}</span>
              <button class="icon-button" type="button" data-
              action="increase" data-id="${item.productId}">+</button>
            </div>
            <button class="icon-button remove-button" type="button"
            data-action="remove" data-id="${item.productId}">Hapus</button>
          </div>
        </li>
      `;
    }).join('');
  }

  function findCartItem(productId) {
    return cart.find(item => item.productId === productId);
  }

  function addToCart(productId) {
    const existingItem = findCartItem(productId);
    if (existingItem) {
      existingItem.quantity += 1;
    } else {
      cart.push({ productId, quantity: 1 });
    }
    renderCart();
  }

  function updateCartItem(productId, change) {
    const existingItem = findCartItem(productId);
    if (!existingItem) return;

    existingItem.quantity += change;
    if (existingItem.quantity <= 0) {
      cart = cart.filter(item => item.productId !== productId);
    }
    renderCart();
  }

```



```
function removeCartItem(productId) {
  cart = cart.filter(item => item.productId !== productId);
  renderCart();
}

function handleProductClick(event) {
  const button = event.target.closest('button');
  if (!button) return;

  const action = button.dataset.action;
  const id = Number(button.dataset.id);

  if (action === 'add') {
    addToCart(id);
  }
}

function handleCartClick(event) {
  const button = event.target.closest('button');
  if (!button) return;

  const action = button.dataset.action;
  const id = Number(button.dataset.id);

  if (action === 'increase') {
    updateCartItem(id, 1);
  }
  if (action === 'decrease') {
    updateCartItem(id, -1);
  }
  if (action === 'remove') {
    removeCartItem(id);
  }
}

function checkout() {
  if (cart.length === 0) {
    alert('Keranjang masih kosong. Tambahkan produk terlebih dahulu.');
```

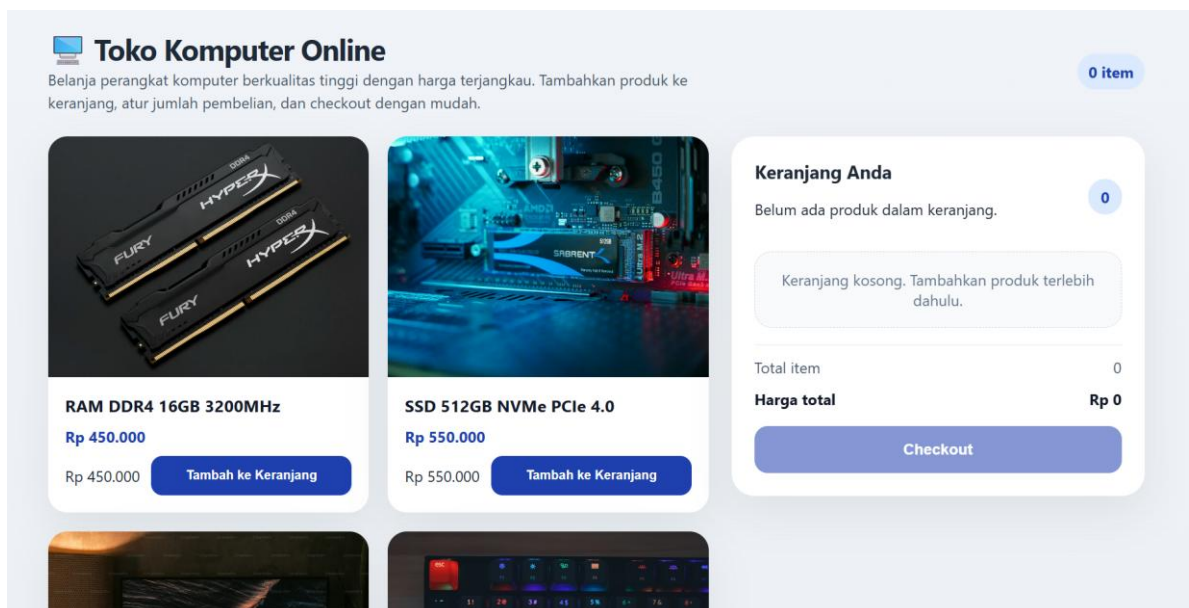
return;

```
  }

  const lines = cart.map(item => {
    const product = products.find(product => product.id === item.productId);
    return `${product.name} x ${item.quantity} = ${formatCurrency(product.price * item.quantity)} `;
  });
}
```

```
const summary = [`Ringkasan Order:`, ...lines, '', `Total Item:`,  
`${getCartItemCount()}`, `Total`, `Harga:`,  
`${formatCurrency(getCartTotal())}`];  
alert(summary.join('\n'));  
}  
  
function init() {  
  renderProducts();  
  renderCart();  
  productGrid.addEventListener('click', handleProductClick);  
  cartList.addEventListener('click', handleCartClick);  
  checkoutButton.addEventListener('click', checkout);  
}  
  
window.addEventListener('DOMContentLoaded', init);
```

Hasil Praktikum: membuat tugas Aplikasi todo list lengkap



BAB IV

PENUTUP

Kesimpulan

Berdasarkan hasil praktikum, dapat disimpulkan bahwa Javascript DOM memungkinkan pengembang untuk mengakses, memanipulasi, dan megubah konten, gaya (CSS), serta element halaman web secara dinamis menggunakan javaScript tanpa perlu memuat ulang halaman.